hw20

In this homework assignment we'll be exploring crop yields in Iowa and investigating whether they correlate with annual precipitation.

0 The data

Crop yields

The crop yield data can be found in the folder "crops". This data is taken from the US Department of Agriculture's Risk Management Agency (https://webapp.rma.usda.gov/apps/RIRS/RMACountyYieldsReport.aspx). Each file in "crops" contains data for all 99 Iowa counties for a single crop in a single year for the years 1990-2025. The key columns we'll be using to investigate the yield are the `Yield.Year` column, which provides the year, and the `Yield.Amount` column, which provides the average yield in bushels per acre.

Precipitation

The precipitation data ("precip.txt") comes from NOAA's nClimDiv (https://www.ncei.noaa.gov/monitoring-content/data/us/climdiv/monthly/current/) and provides monthly estimates of precipitation for all US counties since 1895 (though I've narrowed it down to only Iowa counties). Note that the file "county-readme.txt" provides documentation on this data - you'll want to take a look at that to make sure you understand the data.

1 Read in the data

1.1 *Crop data*

The crop data is contained in the folder "crops" - the data has been separated out by year and crop. Read in all of the data on **corn** yields and combine the tables into a single data frame.

One note to make - when you read in the data, R will interpret `County.Code` as an integer and strip the leading 0's - this will cause problems later on. It turns out that `read.csv()` has a convenient parameter called `colClasses` that can help us out here. Take a look at the documentation and/or do some web searching to learn how to use this parameter to ensure that `County.Code` is read in as a character column.

1.1.1 *Plot Sioux County data*

Create a line plot that shows Sioux County's corn yield by year. Save the plot out to a file.

1.2 *Read in the precipitation data*

Read in the precipitation data. Each row represents an Iowa county in a given year, and each column (except for the first) represents one month of the year. The first column contains several pieces of information - take a look at the documentation of the data for more. When reading in the table, give the columns descriptive names. (*Hint: this is another instance where you'll want to use* `colClasses` *to ensure that the first column is read in as a character column.*)

After you've read in the data, break the first column into multiple columns based on the information provided in the documentation (*hint: one of the string functions we've learned about will be very useful for this*).

1.3 *Create a total precipitation column*

Add a column to the data frame that contains the annual precipitation (i.e. the sum of the precipitation for all months).

2 Analyze the data

2.1 *Join the data*

Now that we've read in the data, let's merge the two datasets. Note that the `County.Code` column of the corn yield data corresponds to the county identifier that you extracted from the first column of the precipitation data.

2.2 *Create plots*

We'll use plots for each county to visually investigate if there is any correlation between the total precipitation and yield. For each county, create a scatter plot that plots the total yield against the total precipitation. Create a folder to store the output and then output a plot for each county (so you should end up with 99 images in the folder).

2.3 *Correlation by county*

Let's calculate the correlation coefficient between total precipitation and corn yield for each county. You can use the `cor()` function to calculate the correlation, and you can use `dplyr`'s grouping/summarizing capabilities to run `cor()` for each county. Write out the resulting data frame to a CSV.

3 Conclusion

Write a few sentences describing your findings. Does there appear to be correlation between total precipitation and corn yield?